Status	Finished
Started	Sunday, 18 October 2026, 6:15 PM
Completed	Sunday, 18 October 2026, 6:25 PM
Duration	10 mins 3 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. The parts are ignored
- c. The parts are merged without sorting
- d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes code execution slower
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It sorts data out of the box efficiently
c. It requires external libraries
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Merge Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are compared element by element
- c. They are combined to form a single sorted list
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Bubble Sort
- c. Merge Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It has no effect on the analysis process
c. It makes the analysis slower
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Selection sort
c. Heap sort
d. Merge sort

The correct answer is:
Merge sort

Question **15**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By using the bubble-up method
c. By breaking the input into smaller parts and merging them
d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them